

Ag 12

11)

$x_1, \dots, x_n \sim \text{Pareto}(\lambda, \sigma)$ $\sigma > 0$ conocido

$$f(x|\lambda, \sigma) = \sigma \lambda^\sigma x^{-\sigma-1} \mathbb{I}\{x \geq \lambda\} \quad \lambda > 0$$

$$L(x|\lambda, \sigma) = \sigma^n \lambda^{n\sigma} (\prod x_i)^{-\sigma-1} \mathbb{I}\{\min\{x_i\} \geq \lambda\}$$

$$= \underbrace{\sigma^n (\prod x_i)^{-\sigma-1}}_{h(x)}$$

$$\underbrace{\lambda^{n\sigma}}_{g(T|\sigma)} \mathbb{I}\{T \geq \lambda\}$$

$$T = \min\{x_i\}$$

$$F_X(x) = \int_{\lambda}^x \sigma \lambda^\sigma x^{-\sigma-1} dx$$

$$= \sigma \lambda^\sigma \frac{x^{-\sigma}}{-\sigma} \bigg|_{x=\lambda}^x = 1 - \lambda^\sigma x^{-\sigma}$$

$$F_T(\tau) = P_r(T \leq \tau) = 1 - P_r(T > \tau) = 1 - [P_r(x_i > \tau)]^n$$

$$= 1 - [1 - F_X(\tau)]^n$$

$$= 1 - \lambda^{n\sigma} \tau^{-n\sigma}$$

$$T \sim \text{Pareto}(\lambda, n\sigma)$$

si $a > \lambda$,

$$P_r(a \leq T \leq b) = 1 - 2$$

$$P_r(a \leq T) = 1 - 2$$

$$P_r(T \leq a) = 2/2$$

$$1 - \lambda^{n\sigma} a^{-n\sigma} = 2/2$$

$$1 - 2 = \left(\frac{\lambda}{a}\right)^{n\sigma}$$

$$\left(1 - \frac{2}{2}\right)^{1/n\sigma} = a$$

$$P_r(T \leq b) = 1 - \frac{2}{3}$$

$$1 - \lambda^{n\sigma} b^{-n\sigma} = 1 - \frac{2}{3}$$

$$\left(\frac{\lambda}{b}\right)^{n\sigma} = \frac{2}{3}$$

$$\frac{\lambda}{b} = \left(\frac{2}{3}\right)^{1/n\sigma}$$

$$b = \left(\frac{2}{3}\right)^{-1/n\sigma} \lambda$$

$$P_r(a \leq T \leq b) = 1 - 2$$

$$P_r\left(\left(1 - \frac{2}{3}\right)^{-1/n\sigma} \lambda \leq T \leq \left(\frac{2}{3}\right)^{-1/n\sigma} \lambda\right) = 1 - 2$$

$$P_r\left(\left(1 - \frac{2}{3}\right)^{-1/n\sigma} \leq \frac{T}{\lambda} \leq \left(\frac{2}{3}\right)^{-1/n\sigma}\right) = 1 - 2$$

$$P_r\left(\left(\frac{2}{3}\right)^{1/n\sigma} \leq \lambda \leq \left(1 - \frac{2}{3}\right)^{1/n\sigma} \lambda\right)$$

P2

Resaca de verosimilitud de p x

$$\left\{ \sigma \in \mathcal{O} : \frac{L(\sigma)}{L(\hat{\sigma})} \geq \frac{p}{100} \right\}, \hat{\sigma} = \hat{\sigma}_{EMV}$$

$$x_1, \dots, x_n \stackrel{iid}{\sim} f(x|\sigma) = \frac{2x}{\sigma} \exp\left\{-\frac{x^2}{\sigma}\right\} \quad x > 0, \sigma > 0$$

$$L(x|\sigma) = \prod_{i=1}^n \frac{2x_i}{\sigma} \exp\left\{-\frac{1}{\sigma} \sum x_i^2\right\}$$

$$l(x|\sigma) = n \log(2) - n \log(\sigma) + \sum \log(x_i) - \frac{1}{\sigma} \sum x_i^2 \quad \frac{d}{d\sigma}$$

$$\frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \frac{\sum x_i^2}{\sigma^2} = 0$$

$$\frac{\sum x_i^2}{\sigma^2} = \frac{n}{\sigma}$$

→

$$\hat{\sigma}_{EMV} = \frac{\sum x_i^2}{n}$$

$$\begin{aligned}
\frac{L(\theta)}{L(\hat{\theta})} &= \frac{\gamma^n \theta^{-n} \prod x_i \exp\left\{-\frac{1}{\theta} \sum x_i\right\}}{\gamma^n \hat{\theta}^{-n} \prod x_i \exp\left\{-\frac{1}{\hat{\theta}} \sum x_i\right\}} \\
&= \left(\frac{\hat{\theta}}{\theta}\right)^n \exp\left\{\sum x_i^2 \left(\frac{1}{\hat{\theta}} - \frac{1}{\theta}\right)\right\} \\
&= \exp\left\{n \log(\hat{\theta}/\theta)\right\} \exp\left\{\sum x_i^2 \left(\frac{1}{\hat{\theta}} - \frac{1}{\theta}\right)\right\} \\
&= \exp\left\{\sum x_i^2 \left(\frac{1}{\hat{\theta}} - \frac{1}{\theta}\right) + n \log\left(\frac{\hat{\theta}}{\theta}\right)\right\} \geq \frac{P}{n} / h
\end{aligned}$$

~~Handwritten scribble~~

$$\sum x_i^2 \left(\frac{1}{\hat{\theta}} - \frac{1}{\theta}\right) + n \log\left(\frac{\hat{\theta}}{\theta}\right) \geq C^*$$

$$\frac{\sum x_i^2}{\hat{\theta}} - \frac{\sum x_i^2}{\theta} + n \log(\hat{\theta}) - n \log(\theta) \geq C^*$$

$$n(1 + \log(\hat{\theta})) - \frac{\sum x_i^2}{\theta} - n \log(\theta) \geq C^*$$

$$-\frac{\hat{\theta}_{ENV}^1}{\theta} n - n \log(\theta) \geq C^* - n(1 + \log(\hat{\theta}))$$

$$2(l(\hat{\theta}) - l(\theta)) \sim \chi^2_1$$

$$\begin{aligned}
2\left(\cancel{n \log(2)} - n \log(\hat{\theta}) + \cancel{\sum \log(x_i)} - \frac{1}{\hat{\theta}} \sum x_i^2 - \cancel{n \log(2)} + n \log(\theta) - \cancel{\sum \log(x_i)} + \frac{1}{\theta} \sum x_i^2\right) \\
2\left\{-n(\log(\hat{\theta}) + 1) + n \log(\theta) + \frac{1}{\theta} \sum x_i^2\right\} \leq \chi^2_{1(1-\alpha)} \leq -2 \log(\theta)
\end{aligned}$$

$$P_{\tau} / P_{\sigma} \approx \ln l_9(\sigma) + \frac{1}{\sigma} \frac{\sum x_i^2}{n} \leq \frac{\tilde{\chi}_1^2(1-\alpha)}{2m} + K(l_9(\sigma)+1)$$

$$l_9(\sigma) + \frac{1}{\sigma} \leq \frac{\chi_1^2(1-\alpha)}{2m} + (l_9(\sigma)+1)$$

$$\frac{1}{\sigma} \exp\left\{-\frac{z^2}{\sigma}\right\} \sim \exp(1/\sigma)$$

$$l_9(\sigma) + \frac{d}{d\sigma} l_9(\sigma) \sigma$$

$$\frac{d}{d\sigma} (\sigma + \sigma) l_9(\sigma) - \sigma$$

$$\sum_{i=1}^m \sigma \sim m \sum x_i^2 \sim \chi^2_{2m} \sim \Gamma_m(m, 1/\sigma)$$

P3 $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$ $\sigma = \{1, 2\}$

$$H_0: \lambda=1 \text{ vs } H_1: \lambda=2$$

$$f(x|\lambda) = \lambda e^{-\lambda x}$$

$$L(x|\lambda) = \lambda^n \exp\{-\lambda \sum x_i\}$$

$$\Lambda(\lambda_0, \lambda_1) = \frac{L(\lambda_0)}{L(\lambda_1)} = \left(\frac{\lambda_0}{\lambda_1}\right)^n \exp\{-\sum x_i(\lambda_0 - \lambda_1)\}$$

Lemma Neyman-Pearson

$$C^* = \{x \in \mathcal{X} : \Lambda(\sigma_0, \sigma_1) \leq K_\alpha\}$$

$$\Lambda(1, 2) = 2^{-n} e^{\sum x_i}$$

$$T_4 \quad B C^*(\sigma_0) = 2$$

$$C^* = \{x \in \mathcal{X} : 2^{-n} e^{\sum x_i} \leq K_\alpha\}$$

4

$$P_{\tau}(X \in C^* | \sigma_0 = 1) = 2$$

$$C^* = \{x \in \mathcal{X} : \sum x_i \leq K\}$$

$$\rightarrow x_i \sim \text{Exp}(1)$$

$$\sum x_i \sim \Gamma_n(n, 1)$$

$$2 \sum x_i \sim \chi^2_{2n}$$

$$\rightarrow C^* = \{x \in \mathcal{X} : 2 \sum x_i \leq \tilde{\chi}_{2n}^2(\alpha)\}$$

Problem

$$B_C(\theta_1) = P\left(\{X \in \chi^2: 2 \sum x_i \leq \chi^2_{2n}(\lambda)\} \mid \lambda=2\right)$$

$$\stackrel{n}{2} P\left(\{X \in \chi^2: 4 \sum x_i \leq 2 \chi^2_{2n}(\lambda)\} \mid \lambda=2\right)$$

$$B_C(\theta_1) = F(2 \chi^2_{2n}(\lambda))$$

$$4 \sum x_i \sim \chi^2_{2n}$$

$$F \sim \chi^2_{2n}$$

$$B_C(\theta_1) = \text{Error Type I}$$

$$1 - B_C(\theta_1=2) = \text{Error Type II}$$